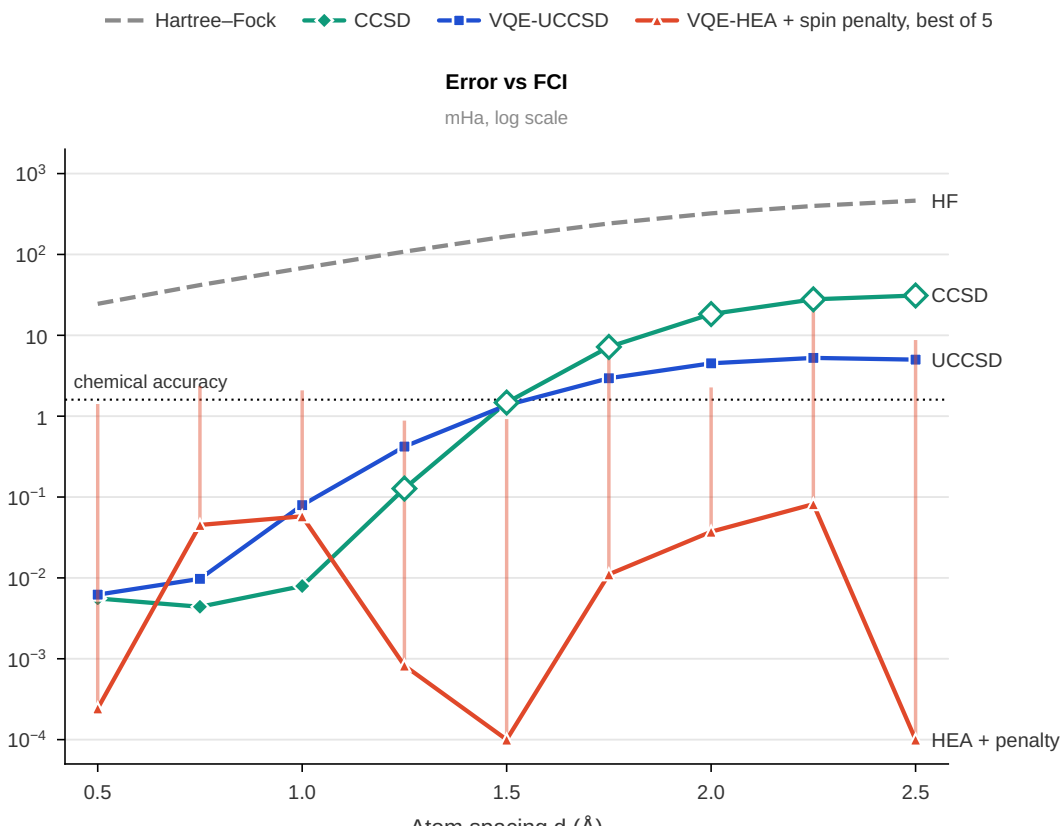
Error vs FCI along the H_4 chain: spin-penalised HEA, UCCSD and CCSD



H_4 chain, STO-3G, noiseless statevector. HEA: 6 layers, cost $E + \mu S^2$ with $\mu = 0.1$ Ha, best of 5 random starts, bars span all 5. Hollow diamonds: CCSD below the exact energy. Errors below 10^{-4} mHa are drawn at the floor.